

HW 06: R Functions

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Section 1 Part A

(A. Can you improve this analysis code?)

```
df <- data.frame(a=1:10, b=seq(200,400,length=10),c=11:20,d=NA) dfa <- -(dfa -  
min(dfa))/(max(dfa) - min(dfa))dfb <- (dfb-min(dfa)) / (max(dfb)-min(dfb)) dfc <- -(dfc  
- min(dfc))/(max(dfc) - min(dfc))dfd <- (dfd - min(dfd)) / (max(dfa) - min(dfd))
```

```
df <- data.frame(a = 1:10,b = seq(200, 400, length = 10),c = 11:20,d = NA)
```

```
rescale01 <- function(x) {  
  if (all(is.na(x))) {return(x)}  
  rng <- range(x, na.rm = TRUE)  
  if(rng[1] == rng[2]) {return(rep(0, length(x)))}  
  (x - rng[1]) / (rng[2] - rng[1])  
}
```

```
df[] <- lapply(df, rescale01)
df
```

```
      a      b      c d
1 0.0000000 0.0000000 0.0000000 NA
2 0.1111111 0.1111111 0.1111111 NA
3 0.2222222 0.2222222 0.2222222 NA
4 0.3333333 0.3333333 0.3333333 NA
5 0.4444444 0.4444444 0.4444444 NA
6 0.5555556 0.5555556 0.5555556 NA
7 0.6666667 0.6666667 0.6666667 NA
8 0.7777778 0.7777778 0.7777778 NA
9 0.8888889 0.8888889 0.8888889 NA
10 1.0000000 1.0000000 1.0000000 NA
```

Section 1 Part B

(B. Can you improve this analysis code?)

```
library(bio3d) s1 <- read.pdb("4AKE") # kinase with drug s2 <- read.pdb("1AKE") #
kinase no drug s3 <- read.pdb("1E4Y") # kinase with drug s1.chainA <- trim.pdb(s1,
chain="A", eley="CA") s2.chainA <- trim.pdb(s2, chain="A", eley="CA") s3.chainA
<- trim.pdb(s3, chain="A", eley="CA") s1.b <- s1.chainA$atom$b s2.b <- s2.chainA$atom$b
s3.b <- s3.chainA$atom$b plotb3(s1.b, sse=s1.chainA, typ="l", ylab="Bfactor") plotb3(s2.b,
sse=s2.chainA, typ="l", ylab="Bfactor") plotb3(s3.b, sse=s3.chainA, typ="l", ylab="Bfac-
tor")
```

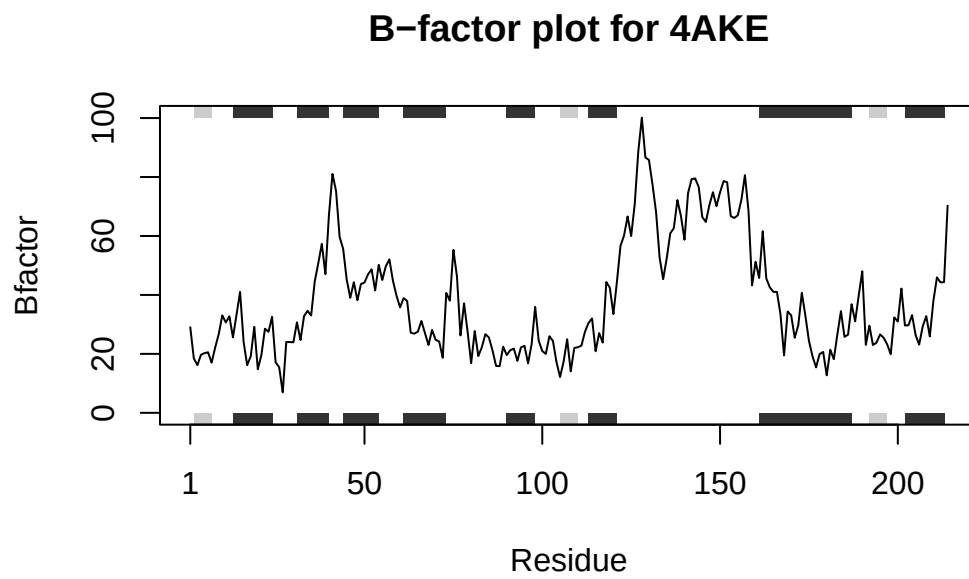
```
library(bio3d)
```

```
analyze_pdb <- function(pdb_id, chain = "A", eley = "CA") {
  pdb <- read.pdb(pdb_id)
  pdb_chain <- trim.pdb(pdb, chain = chain, eley = eley)
  bvals <- pdb_chain$atom$b
  plotb3(bvals, sse = pdb_chain, typ = "l", ylab = "Bfactor",
  main = paste("B-factor plot for", pdb_id))
  return(bvals)}

```

```
s1.b <- analyze_pdb("4AKE")
```

Note: Accessing on-line PDB file

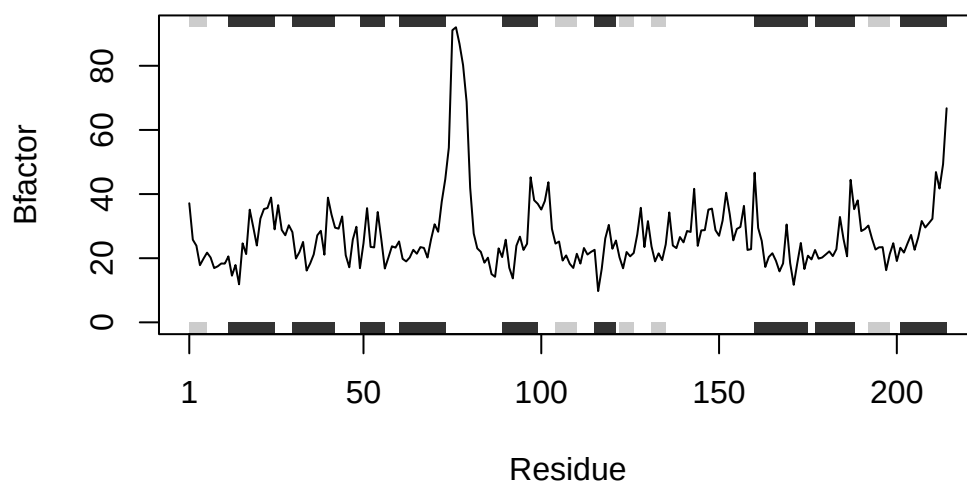


```
s2.b <- analyze_pdb("1AKE")
```

Note: Accessing on-line PDB file

PDB has ALT records, taking A only, `rm.alt=TRUE`

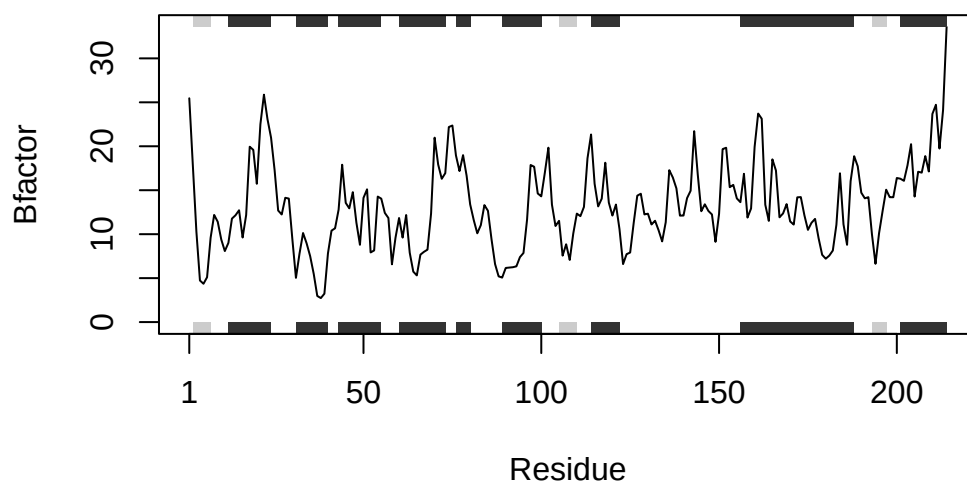
B-factor plot for 1AKE



```
s3.b <- analyze_pdb("1E4Y")
```

Note: Accessing on-line PDB file

B-factor plot for 1E4Y



Questions 1-6

Q1. What type of object is returned from the `read.pdb()` function?

`read.pdb()` returns an object of class `pdb`, which is an object containing atom records, coordinates, residue information, and other structural data.

Q2. What does the `trim.pdb()` function do?

`trim.pdb()` creates a new `pdb` object that contains only a selected subset of atoms such as one chain.

Q3. What input parameter would turn off the marginal black and grey rectangles in the plots and what do they represent in this case?

The rectangles are the secondary structure element annotations, which represent features such as alpha helices and beta strands for the trimmed protein chain.

Q4. What would be a better plot to compare across the different proteins?

A better comparison would be a single overlay plot with all three B-factor profiles on the same axes since it lets you directly compare trends between residues.

Q5. Which proteins are more similar to each other in their B-factor trends. How could you quantify this? **HINT:** try the `rbind()`, `dist()` and `hclust()` functions together with a resulting dendrogram plot. Look up the documentation to see what each of these functions does.

The two proteins that cluster together first in the dendrogram are the most similar in their B-factor trends. One can quantify similarity by combining the B-factor vectors with `rbind()`, calculating pairwise distances with `dist()`, and clustering them with `hclust()`.

Q6. How would you generalize the original code above to work with any set of input protein structures?

You generalize it by writing a function that takes any PDB ID, reads the structure, trims the selected chain and atom type, extracts the B-factors, and makes the plot.